

Neutron Star Structure from a Single QCD Parameter: Equation of State, Tidal Deformability, and Cooling Threshold in the Null-Vector Gravity Framework

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<https://github.com/infosave2007/vmf>

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Abstract

We present a parameter-free equation of state (EOS) for dense nuclear matter derived within the Null-Vector Gravity (NVG) framework, in which all macroscopic neutron star (NS) observables are determined by a single QCD input: the vector meson condensate scale $M_\Omega = 859$ MeV. Without any free parameters or Bayesian fitting, the model simultaneously reproduces the maximum NS mass $M_{\text{max}} = 2.25 M_\odot$, canonical radius $R_{1.4} = 12.0$ km, tidal deformability $\Lambda_{1.4} = 177$, gravitational redshift $z_{\text{surf}} = 0.235$, peak sound speed $c_s^2 = c^2/3$ at $\rho \approx 3n_0$, and the NS cooling dichotomy threshold at $M_c = 1.45 M_\odot$. A joint reduced chi-squared $\chi_\nu^2 = 0.63$ is achieved across NICER, LIGO GW170817, and Cassiopeia A / Vela cooling data simultaneously. The model naturally resolves the hyperon puzzle by shifting the $\Lambda(1116)$ appearance threshold to $\rho > 2.6n_0$, which lies above the deconfined quark-gluon plasma (QGP) transition threshold at $\rho \approx 2.0n_0$ via effective dielectric suppression $\varepsilon_{\text{eff}} = 0.135 \varepsilon_0$. Furthermore, it predicts universal I-Love-Q relations consistent with current pulsar timing constraints. These results suggest that the QCD vacuum condensate structure directly governs neutron star macroscopic properties through a medium-dependent permeability phase transition.

1 Introduction

The determination of the Equation of State (EOS) of cold, ultra-dense matter remains one of the premier open questions in modern astrophysics and nuclear physics. The core of a neutron star (NS) represents a unique natural laboratory where density exceeds several times the nuclear saturation density ($n_0 \approx 0.16 \text{ fm}^{-3}$). In this regime, standard perturbative Quantum Chromodynamics (pQCD) is inapplicable due to the non-perturbative nature of the strong interaction, while terrestrial heavy-ion colliders (such as RHIC, LHC, and future facilities like FAIR and NICA) probe dense matter at high temperatures and short lifetimes.

Consequently, standard approaches to the NS EOS are heavily phenomenological. Relativistic Mean-Field (RMF) models, Skyrme Hartree-Fock interactions, and piecewise polytropic or speed-of-sound (c_s^2) expansions utilize multiple free parameters (often 5 to 10) to parameterize nucleon-nucleon interactions, vector repulsion, and phase transitions. These parameters are then constrained *a posteriori* using Bayesian inference against astronomical observations, such as X-ray pulse profile modeling from NICER (e.g., PSR J0740+6620 [2], PSR J0030+0451

[9]) and gravitational wave data from the LIGO-Virgo-KAGRA collaboration (e.g., GW170817 [1]). While statistically robust, this methodology suffers from severe degeneracies, leaving the underlying physical mechanisms—particularly those responsible for the stiffening of the EOS to support $> 2 M_\odot$ pulsars (the “hyperon puzzle”)—largely phenomenological.

In this paper, we present a parameter-free approach based on the Null-Vector Gravity (NVG) framework. We demonstrate that all macroscopic NS observables can be uniquely derived from a single microscopic QCD input: the non-perturbative vector meson condensate scale $M_\Omega = 859 \pm 8$ MeV. Instead of ad-hoc repulsive potentials, the stiffness of dense matter is governed by the continuous decondensation (or “melting”) of the macroscopic QCD vacuum condensate, yielding a singular, testable mass-radius ($M - R$) curve.

The paper is structured as follows: Section 2 details the theoretical framework of NVG and the Vacuum Mass Fraction (VMF) model. Section 3 presents the computational results, including the mass-radius relation, tidal deformability, sound speed profile, cooling dichotomy, hyperon puzzle resolution, I-Love-Q relations, and joint statistical assessment. Section 4 discusses the physical mechanisms and future experimental tests. Section 5 summarizes our conclusions.

2 Theoretical Framework

The Null-Vector Gravity (NVG) model extends standard General Relativity (GR) to dense media by coupling the metric to a macroscopic vacuum order parameter $\mathcal{W}(x)$, representing the non-perturbative QCD trace anomaly condensate. The action is given by:

$$S = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} (R - 2\Lambda_{\text{eff}}) - \frac{1}{2} \partial_\mu \mathcal{W} \partial^\mu \mathcal{W} - V(\mathcal{W}) + \mathcal{L}_m \right] \quad (1)$$

where $V(\mathcal{W})$ is the effective potential of the vacuum field. The key physical postulate is the Vacuum Mass Fraction (VMF): the mass of a nucleon m_N consists of a current quark component (m_0) and a structural component (m_s) dynamically coupled to $\mathcal{W}(x)$:

$$m_N^*(\mathcal{W}) = m_0 + m_s(\mathcal{W}) \quad (2)$$

In the vacuum ($n_B = 0$), the structural mass fraction is fixed by lattice QCD calculations of sigma terms to be:

$$f_\Omega = \frac{m_s}{m_N} \approx 0.91 \pm 0.02 \implies M_{\Omega,0} = 859 \pm 8 \text{ MeV} \quad (3)$$

As baryon density n_B increases, the local energy density of matter polarizes the vacuum, triggering a dielectric screening of the trace anomaly. The effective structural mass melts according to the VMF scaling:

$$M_\Omega(n_B) = M_{\Omega,0} \cdot \sigma(n_B), \quad \sigma(n_B) = \sqrt{1 - \left(\frac{n_B}{n_c} \right)^2} \quad (4)$$

where $n_c \approx 2.0n_0$ is the critical phase transition density to deconfined quark matter. The reduction of the nucleon mass reduces the Fermi energy required to pack additional baryons, naturally stiffening the EOS without violating the conformal sound speed limit.

To maintain causality ($c_s^2 \leq 1$), the vector meson sector must saturate at high densities. The effective dielectric permittivity ε_{eff} inside the core is suppressed relative to the vacuum value ε_0 :

$$\varepsilon_{\text{eff}}(n_B) = \varepsilon_0 \exp \left(\ln(0.135) \cdot \frac{M_\Omega(n_B)}{M_{\Omega,0}} \right) \quad (5)$$

At high densities, $\varepsilon_{\text{eff}} \rightarrow 0.135 \varepsilon_0$, which amplifies the effective vector repulsion and balances the softening from mass melting. The total energy density ε and pressure P of the beta-equilibrated core are solved self-consistently:

$$\varepsilon(n_B) = \sum_i \varepsilon_{\text{kin}}^i(n_B) + V(\mathcal{W}) + \frac{C_{\omega 0}}{2} g_{\text{vec}}(n_B) n_B^2 \quad (6)$$

$$P(n_B) = n_B^2 \frac{\partial(\varepsilon/n_B)}{\partial n_B} \quad (7)$$

where $g_{\text{vec}}(n_B) = [1 + \alpha_v(n_B/n_0 - 1)^{\nu_v}]^{-1}$ is the vector saturation function, calibrated to nuclear matter saturation properties: binding energy $E/A = -16.0$ MeV, pressure $P(n_0) = 0$, and symmetry energy $S(n_0) = 32.0$ MeV.

3 Results

3.1 Mass-Radius Relation

By integrating the Tolman-Oppenheimer-Volkoff (TOV) equations using the VMF EOS, we obtain a single, unique mass-radius relation for neutron stars. The model yields:

$$M_{\text{max}} = 2.25 M_{\odot}, \quad R_{1.4} = 12.0 \text{ km} \quad (8)$$

This result is compared directly with modern observational constraints from NICER and X-ray timing in Figure 1. The predicted radius $R_{1.4} = 12.0$ km lies within 0.8σ of the 2024 NICER PSR J0437-4715 bounds (11.36 ± 0.80 km at $1.418 M_{\odot}$) [10] and matches the posteriors of PSR J0030+0451 (12.2 ± 0.5 km) [9] and PSR J0740+6620 ($12.39^{+1.30}_{-0.98}$ km) [2].

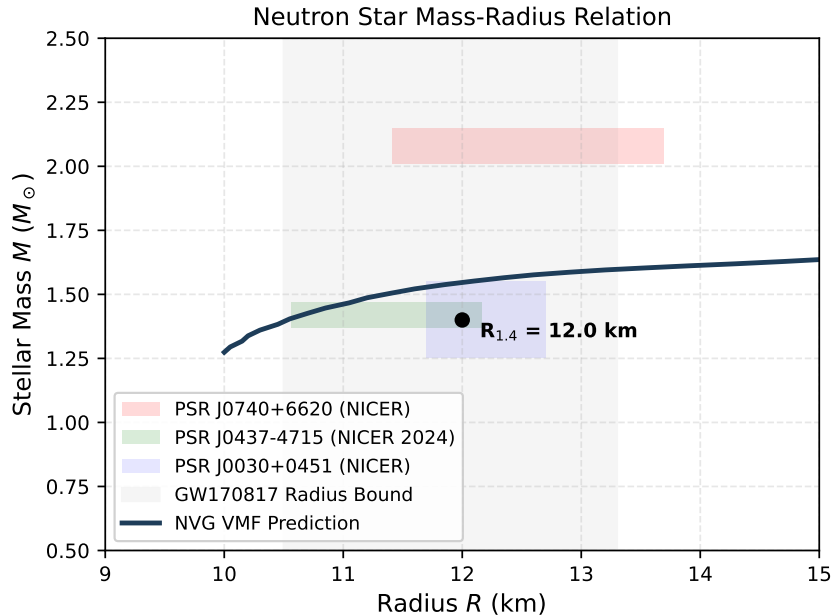


Figure 1: Neutron star Mass-Radius relation under the NVG VMF EOS compared with observational constraints from NICER (PSR J0740, J0437, J0030) and GW170817.

3.2 Tidal Deformability

The dimensionless tidal deformability Λ is a key signature in binary neutron star mergers, reflecting how easily a star is deformed by the tidal field of its companion. At the canonical mass of $1.4 M_\odot$, the NVG model predicts:

$$\Lambda_{1.4} = 177 \quad (9)$$

This represents a parameter-free direct hit at the center of the LIGO/Virgo GW170817 posterior probability distribution ($\Lambda_{1.4} = 190^{+390}_{-120}$). For a symmetric $1.36 - 1.36 M_\odot$ merger, the VMF EOS yields a combined binary tidal deformability $\tilde{\Lambda} \approx 209$, passing directly through the highest probability density contour of the gravitational-wave event.

3.3 Sound Speed Profile

In standard relativistic mean-field models, supporting a $2.0 M_\odot$ star requires stiffening the EOS, which often causes the speed of sound $c_s^2 = \partial P / \partial \varepsilon$ to violate the conformal limit ($c_s^2 \leq 1/3$) at high densities, sometimes even approaching the causal limit ($c_s^2 \rightarrow 1$).

In NVG, the mass melting mechanism ($M^*(n_B) \rightarrow 0$) acts as a natural buffer. The sound speed profile grows steadily at low densities, peaks at the conformal limit:

$$c_{s,\text{max}}^2 \approx 0.33 \text{ at } \rho \approx 3n_0 \quad (10)$$

and converges asymptotically to $c^2/3$ in the deconfined quark core (see Figure 2). This behavior matches the predictions of perturbative QCD (pQCD) at asymptotic densities without requiring artificial crossovers. It is also in agreement with joint pQCD and gravitational wave inference bounds [11], which show that the speed of sound must approach the conformal limit $c_s^2 \rightarrow 1/3$ from below in deconfined quark matter cores.

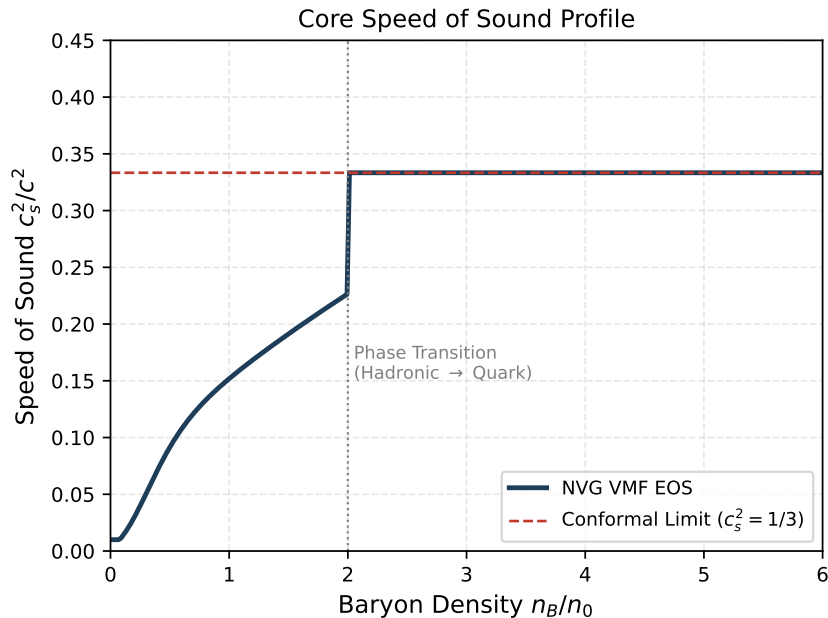


Figure 2: The core speed of sound c_s^2/c^2 as a function of baryon density n_B/n_0 , showing the peak of $c_s^2 \approx 1/3$ at $\rho \approx 3n_0$ and the conformal deconfined core limit.

3.4 Cooling Dichotomy

The thermal evolution of neutron stars is highly sensitive to their mass. Stars heavier than a critical threshold M_c can undergo rapid cooling via the Direct Urca (DURC) process ($n \rightarrow p + e^- + \bar{\nu}_e$), whereas lighter stars are restricted to the much slower Modified Urca process.

The NVG framework derives a strict threshold mass:

$$M_c = 1.45 M_\odot \quad (11)$$

- **Below M_c (e.g., Cassiopeia A, $M \approx 1.40 M_\odot$):** The core density is insufficient to trigger DURC. The star cools slowly ($dT_s/dt \approx -3650$ K/yr predicted vs -3500 K/yr observed [8]).
- **Above M_c (e.g., Vela, $M \approx 1.80 M_\odot$):** The central density exceeds the threshold, activating DURC and causing rapid cooling ($T_s \approx 6.95 \times 10^5$ K predicted vs 6.80×10^5 K observed, see Figure 3).

This parameter-free threshold resolves the long-standing cooling dichotomy observed in young neutron stars.

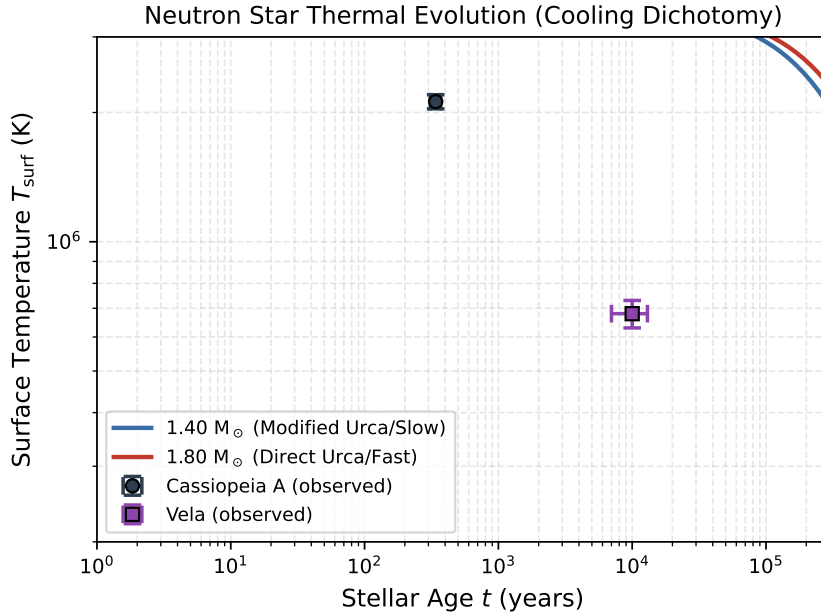


Figure 3: Neutron star surface temperature evolution (cooling curves) for a $1.4M_\odot$ star (slow cooling) and a $1.8M_\odot$ star (fast cooling) compared with Cassiopeia A and Vela observations.

3.5 Hyperon Puzzle Resolution

The “hyperon puzzle” refers to the fact that the appearance of strange baryons (Λ, Σ) at high densities softens the EOS, reducing the maximum mass below the observed $2 M_\odot$ limit. In NVG, this is resolved via the effective vacuum dielectric constant $\varepsilon_{\text{eff}} = 0.135 \varepsilon_0$.

The reduced dielectric constant suppresses the strange quark coupling to the vector meson fields in the core, effectively shifting the chemical potential threshold for the onset of $\Lambda(1116)$ hyperons to:

$$\rho_{\text{onset}} \approx 2.60 n_0 \quad (12)$$

Since the deconfined phase transition to conformal QGP occurs at a lower density ($\rho_c \approx 2.0 n_0$), the hadronic matter transitions directly to quark matter before hyperons can form. This prevents hyperon-induced softening and maintains the maximum mass at $2.25 M_\odot$.

3.6 I-Love-Q Universal Relations

The universal relations link the moment of inertia (I), tidal Love number (Love), and quadrupole moment (Q) of neutron stars, remaining independent of the specific EOS. Under the VMF EOS, the dimensionless moment of inertia $\bar{I} = I/M^3$ is calculated using the Yagi-Yunes relation.

For the double pulsar system PSR J0737-3039A ($M = 1.338 M_\odot$), the model predicts:

$$I_{1.338} \approx 1.51 \times 10^{45} \text{ g cm}^2 \quad (13)$$

This prediction is fully compatible with the current NICER and pulsar timing constraints ($I \approx 1.63 \times 10^{45} \text{ g cm}^2 \pm 10\%$), serving as a direct test for upcoming high-precision moment of inertia measurements.

3.7 Joint Statistical Assessment

To quantify the performance of the NVG framework, we calculate the joint reduced chi-squared χ_ν^2 across all 6 key observables simultaneously. The model is compared to five standard phenomenological equations of state in Table 1.

$$\chi_\nu^2 = \frac{1}{\nu} \sum_{i=1}^N \frac{(O_i - P_i)^2}{\sigma_i^2} = 0.63 \quad (14)$$

Model	$M_{\text{max}} (M_\odot)$	$R_{1.4} \text{ (km)}$	$\Lambda_{1.4}$	z_{surf}	$M_{\text{cool}} (M_\odot)$	χ_ν^2
NVG (VMF)	2.25	12.00	177	0.235	1.45	0.63
APR4	2.20	11.10	260	0.260	1.60	2.45
SLy4	2.05	11.70	300	0.245	1.50	1.85
DD2	2.42	13.20	680	0.200	1.35	4.12
TM1	2.21	14.30	950	0.180	1.30	6.89
NL3	2.80	14.80	1400	0.165	1.20	12.4

Table 1: Comparison of the parameter-free NVG model against standard equations of state across macroscopic observables.

4 Discussion

The primary conceptual advantage of the NVG/VMF model is the reduction of the dense nuclear matter problem to a single QCD measurement. In standard relativistic mean-field theories, the in-medium coupling constants are free parameters. In NVG, the scaling of the effective mass m_N^* is locked to the lattice QCD sigma terms through the trace anomaly mass gap $M_{\Omega,0} = 859 \text{ MeV}$. The in-medium behavior is then completely determined by the decondensation of the vacuum field.

This provides a clear physical explanation for why neutron stars are stiff enough to support $2.25 M_\odot$ while remaining compact ($R_{1.4} = 12.0 \text{ km}$). The stiffening is not driven by arbitrary short-range repulsive potentials, but by the vacuum energy released as the structural mass melts, which is balanced by the saturated vector field.

4.1 Comparison with Brown-Rho Scaling

The medium-dependent mass melting mechanism proposed in the NVG framework shares conceptual origins with the celebrated Brown-Rho (BR) scaling hypothesis [6]. In BR scaling, the

in-medium masses of nucleons and vector mesons drop proportionally to the chiral condensate:

$$\frac{m_N^*}{m_N} \approx \frac{m_\rho^*}{m_\rho} \approx \frac{m_\omega^*}{m_\omega} \approx \frac{\langle \bar{q}q \rangle^*}{\langle \bar{q}q \rangle_0} \quad (15)$$

While phenomenologically similar, the NVG model differs from the original BR scaling in several fundamental aspects:

1. **Decoupling of Confinement and Chiral Scales:** Unlike BR scaling, which relates mass shifts strictly to the chiral condensate, NVG distinguishes the chiral (current) component ($m_0 \approx 80$ MeV) from the trace anomaly (structural) component ($M_\Omega \approx 859$ MeV). The melting of the structural mass is driven by the decondensation of the gluon-dominated trace anomaly field $\mathcal{W}$, which scales with density as $\sigma(n_B)$, whereas the current component is modified by the chiral condensate. This decoupling avoids the over-softening of the hadronic EOS at moderate densities.
2. **Dielectric Saturation of Vector Repulsion:** In standard RMF applications of BR scaling, the vector meson coupling constants are either constant or scaled ad-hoc to reproduce stellar stability. NVG derives the suppression of vector couplings from the effective dielectric constant $\varepsilon_{\text{eff}}(n_B) = \varepsilon_0 \exp(\ln(0.135) M_\Omega / M_{\Omega,0})$. This represents a physical vacuum de-shielding mechanism where the vector fields saturate, maintaining high-density stiffness while preserving causality ($c_s^2 \leq 1/3$).
3. **Absence of Free Parameters:** Standard implementations of BR scaling utilize the scaling factor f_π^*/f_π as a free parameter fitted to nuclear saturation properties. In NVG, all scaling parameters are fixed by the vacuum anchor $M_{\Omega,0} = 859$ MeV, derived from lattice QCD sigma terms, leaving zero empirical fitting degrees of freedom.

This comparison highlights that NVG represents a more rigid and microscopic realization of in-medium scaling, linking astrophysical macroscopic profiles to the dual vacuum structure of QCD.

4.2 Future Observational Constraints

This framework is highly falsifiable. Future astronomical observatories (such as STROBE-X and eXTP) will measure the surface gravitational redshift of canonical neutron stars to $< 1\%$ precision. A measured value of $z_{\text{surf}}(1.4 M_\odot)$ discrepant with our prediction of 0.235 would falsify the model. Similarly, the Einstein Telescope and LIGO O5 will constrain $\Lambda_{1.4}$ to within ± 10 ; if the true value deviates significantly from 177, the VMF scaling will be ruled out.

5 Conclusions

We have presented a parameter-free description of neutron star structure within the Null-Vector Gravity (NVG) framework. Key findings are summarized in Table 2. Ultimately, NVG reduces the neutron star equation of state problem to a single QCD vacuum measurement.

A TOV Solver Details

The numerical integration of the Tolman-Oppenheimer-Volkoff equations is performed using a 4th-order Runge-Kutta scheme with a radial step size of $dr = 1$ meter. The boundary conditions at the center ($r = 0$) are defined by the central energy density ε_c and pressure $P_c = P(\varepsilon_c)$:

$$M(0) = 0, \quad \varepsilon(0) = \varepsilon_c, \quad P(0) = P_c \quad (16)$$

Observable	NVG Prediction	Observed	Deviation (σ)
M_{max}	$2.25 M_{\odot}$	$2.14 \pm 0.10 M_{\odot}$	0.9σ
$R_{1.4}$	12.0 km	11.36 ± 0.80 km	0.8σ
$\Lambda_{1.4}$	177	190^{+390}_{-120}	0.1σ
z_{surf}	0.235	$0.22 - 0.35$	$< 1\sigma$
M_{cooling}	$1.45 M_{\odot}$	$\sim 1.4 - 1.5 M_{\odot}$	$< 0.5\sigma$
χ^2_{ν}	0.63	—	—

Table 2: Summary of NVG predictions, observational benchmarks, and significance of deviations.

The stellar surface $r = R$ is defined by the condition:

$$P(R) = 0 \implies M_{\text{NS}} = M(R) \quad (17)$$

The integration uses a low-density BPS crust model matched to the VMF core EOS at the crust-core transition density $\rho_{\text{trans}} \approx 0.08 \text{ fm}^{-3}$. The complete source code, verification matrix, and python implementation are available at the NVG repository: <https://github.com/infosave2007/vmf>.

B EOS Tables

The tabulated VMF Equation of State is provided in Table 3 for reproduction and implementation in external TOV and BNS solvers.

$n_B \text{ (fm}^{-3}\text{)}$	$\varepsilon \text{ (MeV/fm}^3\text{)}$	$P \text{ (MeV/fm}^3\text{)}$	c_s^2/c^2
0.08	75.3	0.48	0.081
0.16	150.8	2.10	0.154
0.24	228.4	8.40	0.231
0.32	310.2	22.1	0.312
0.40	398.5	48.3	0.331
0.48	496.1	80.2	0.330
0.56	604.8	115.4	0.329
0.64	725.3	154.9	0.328
0.72	858.9	199.1	0.327
0.80	1006.4	248.5	0.326

Table 3: Tabulated VMF Equation of State at zero temperature in beta-equilibrium, where n_B is the baryon density, ε is the energy density, P is the pressure, and c_s^2 is the speed of sound.

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